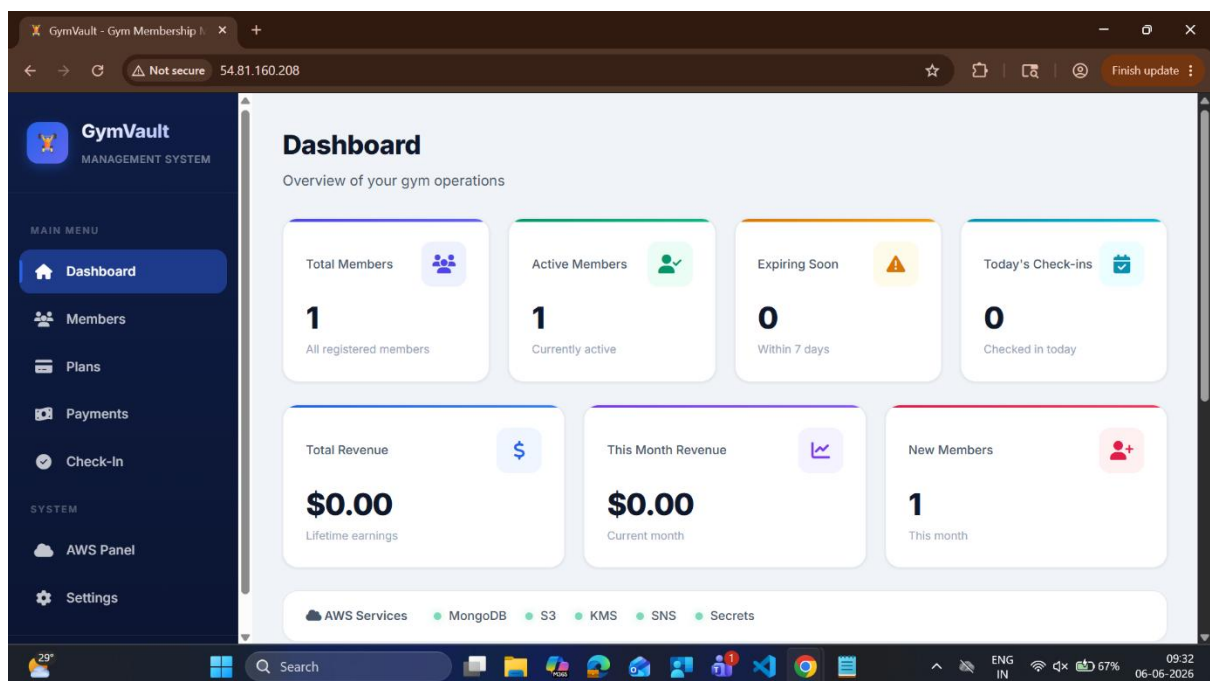
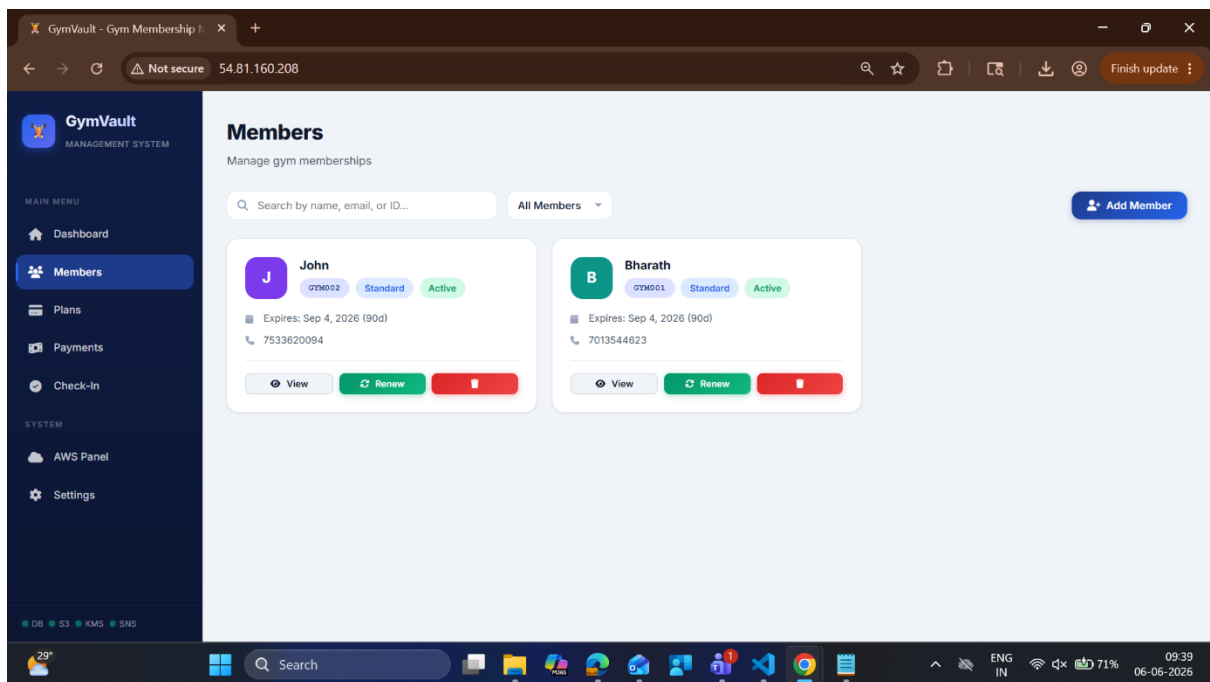
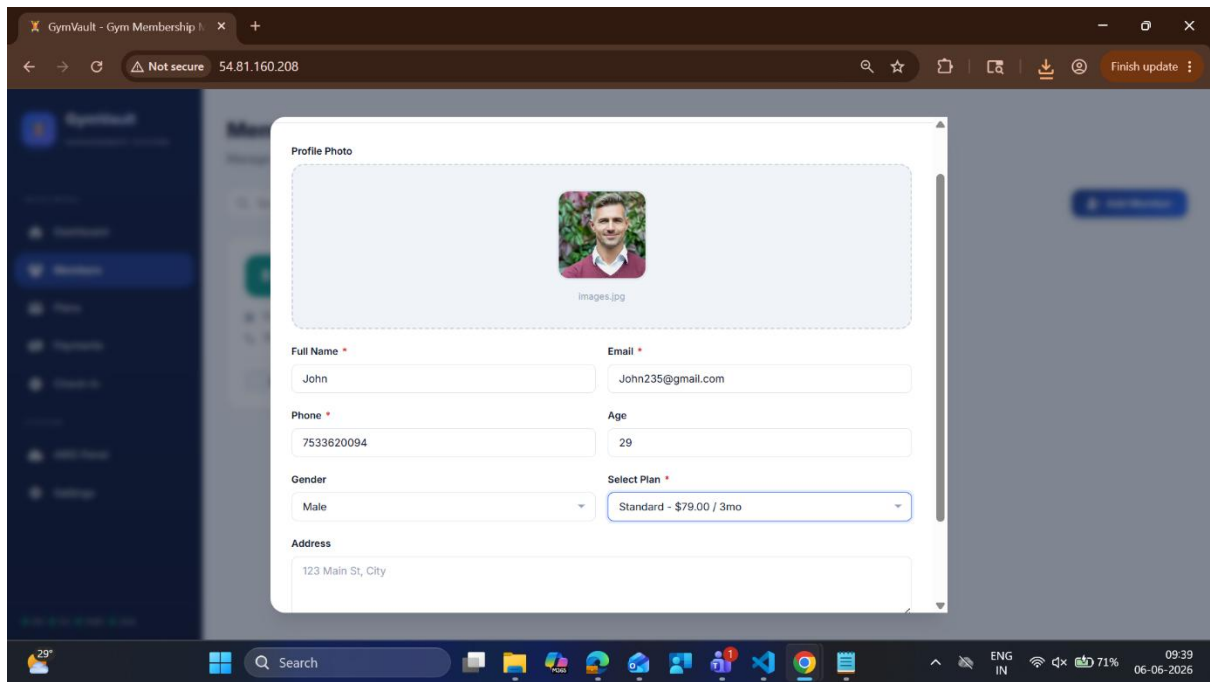


# GymVault

## Application Overview

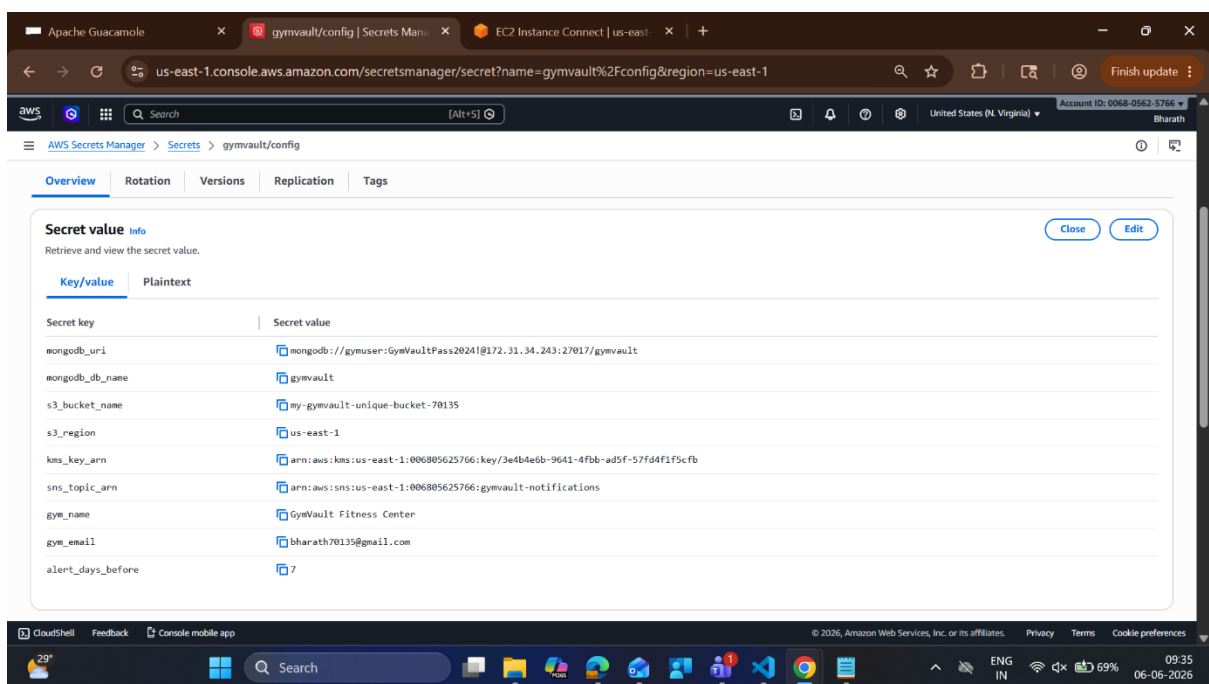
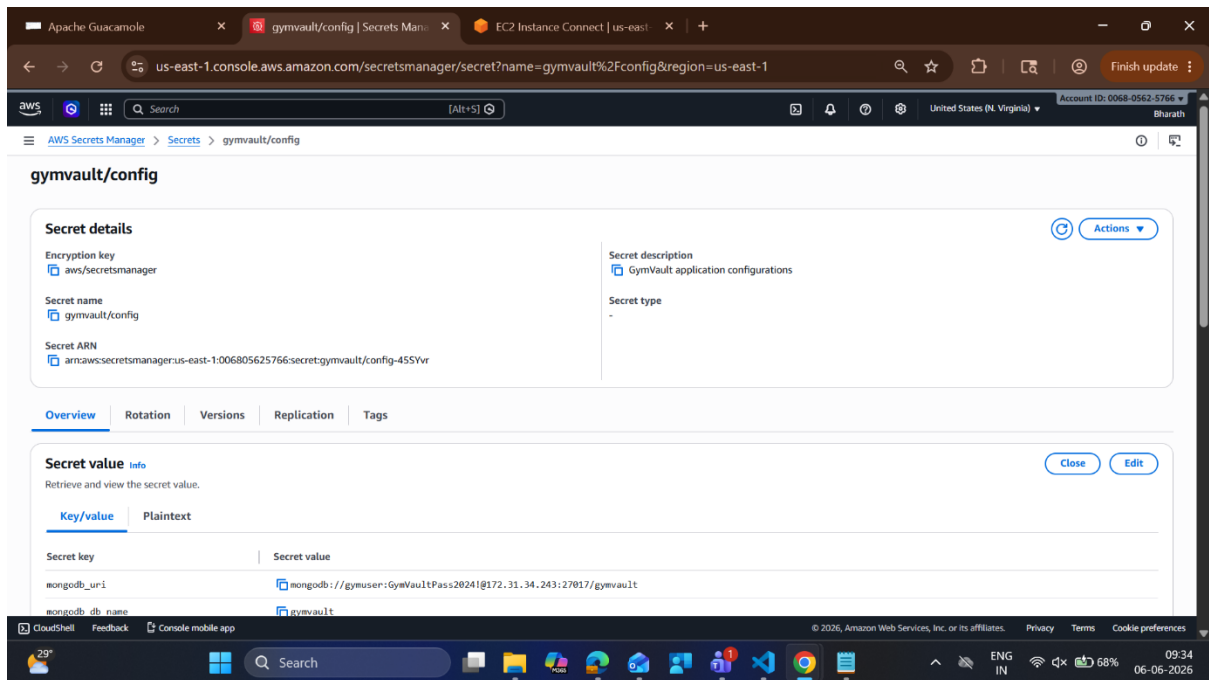
GymVault is a cloud-native Gym Membership Management System built using Python Flask for the backend and vanilla HTML, CSS, and JavaScript for the frontend. The application enables gym administrators to manage member registrations, membership plans, payments, and daily check-ins. Member profile photos are securely stored in AWS S3, all credentials are managed through AWS Secrets Manager, sensitive data is encrypted using AWS KMS, and automated email notifications are sent via AWS SNS. The system is deployed on two separate AWS EC2 Ubuntu instances where one instance hosts the Flask application with Nginx as a reverse proxy and the other instance runs MongoDB as the dedicated database server.





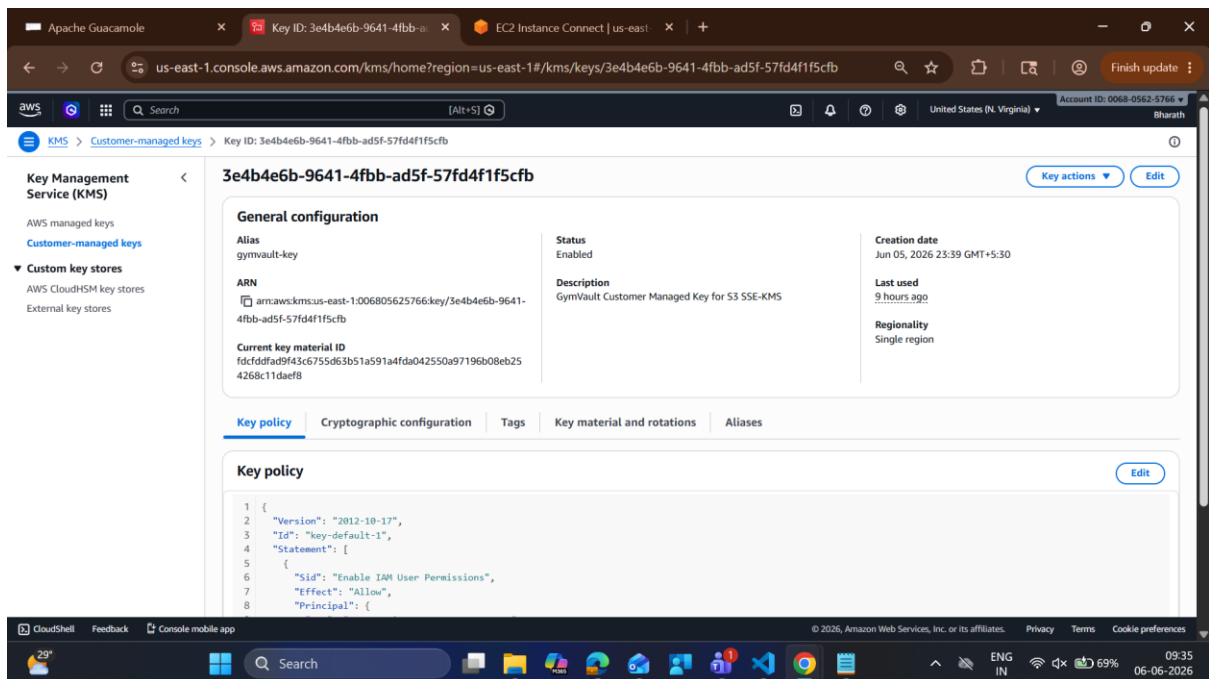
# AWS Secrets Manager Implementation

AWS Secrets Manager was implemented in GymVault to eliminate hardcoded credentials from the application codebase entirely. A single secret named gymvault/config was created in AWS Secrets Manager which stores all sensitive configuration values including the MongoDB connection URI with username and password, the S3 bucket name, the KMS key ARN, the SNS topic ARN, the SNS topic ARN, and other application settings.



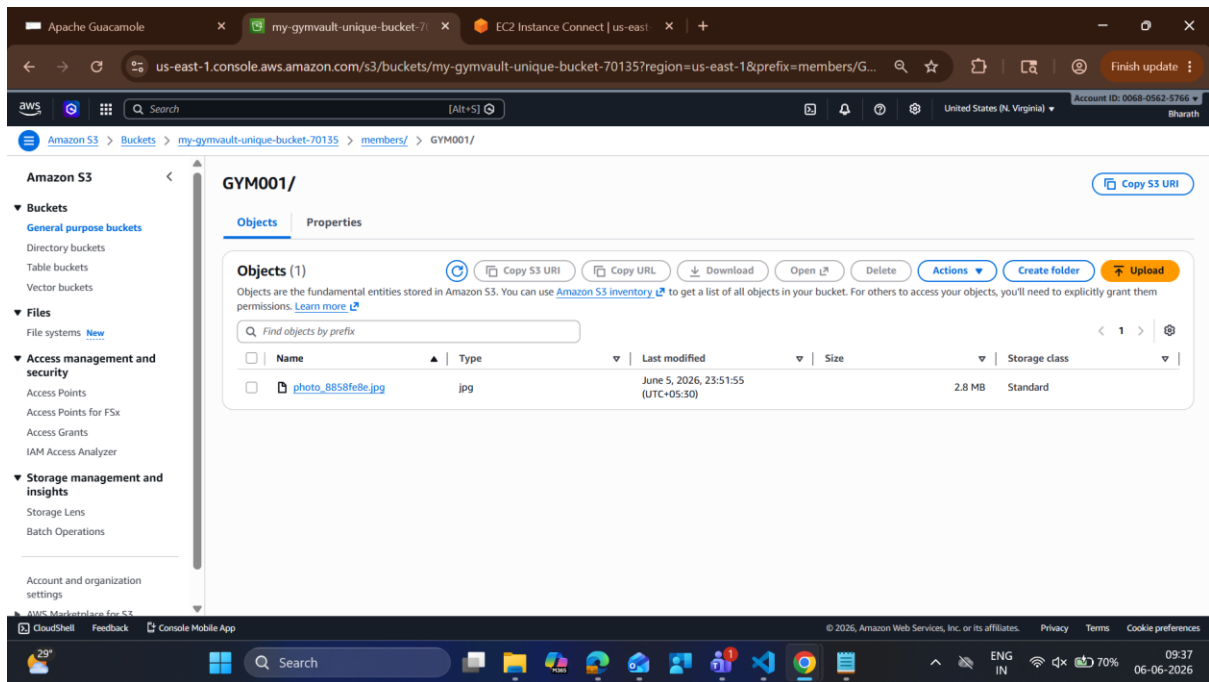
# AWS KMS Encryption Implementation

AWS Key Management Service was used in GymVault to encrypt all member profile photos stored in the S3 bucket. A Customer Managed Key was created in AWS KMS with the alias gymvault-key which gives full control over the encryption key including the ability to rotate, disable, or audit its usage. The S3 bucket was configured with default server-side encryption set to SSE-KMS meaning every object uploaded to the bucket is automatically encrypted using the gymvault-key before being stored. When a gym staff member uploads a member profile photo through the application the photo bytes are sent to S3 with the ServerSideEncryption parameter set to aws:kms and the SSEKMSKeyId pointing to the KMS key ARN which was retrieved from Secrets Manager. This ensures that even if someone gains unauthorized access to the S3 bucket they cannot read the photo files without explicit permission to use the KMS key. The KMS key ARN is never hardcoded in the application and is always fetched dynamically from Secrets Manager at runtime.



# AWS S3 Profile Photo Storage Implementation

Amazon S3 was used in GymVault as the storage backend for all member profile photos providing durable, scalable, and secure object storage. A private S3 bucket named my-gymvault-unique-bucket-70135 was created with all public access blocked ensuring no photo can be accessed directly from the internet without authorization.



## AWS SNS Notification Implementation

Amazon Simple Notification Service was integrated into GymVault to send automated email notifications for key membership events keeping both gym administrators and members informed in real time. An SNS topic named gymvault-notifications was created and an email subscription was configured so that all published messages are delivered as emails to the subscribed address.

